Domaine Sciences et Technologies
MASTER 1 CMB

Aix*Marseille université

Graph Theory and Algorithms : TP
Code UE: SCMB02AL

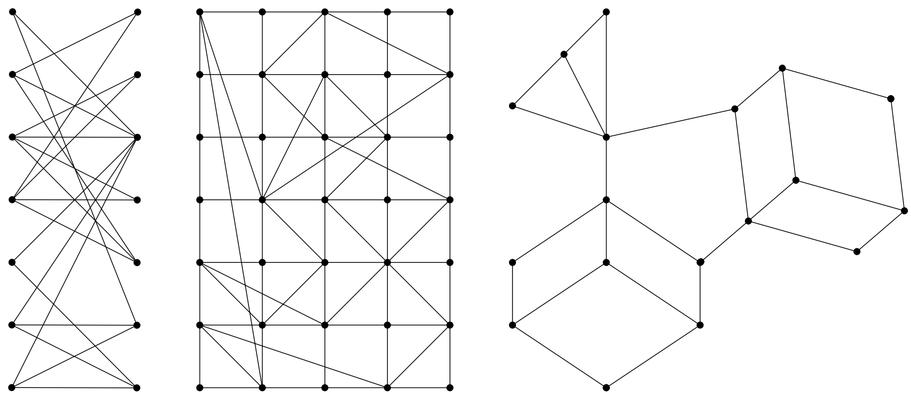
Année 2024-2025

Network Analysis

The goal of this homework is to analyse some graphs using one of the Python packages NetworkX or igraph for creation, manipulation, and study of complex networks. The tutorials on these packages can be found on the links https ://igraph.org/python/tutorial/0.9.8/analysis.html and https ://networkx.org/documentation/stable/tutorial.html. The choice of the package is by you. The deadline is March 24, 2025.

1 What do you have to do ?

You have to analyse 3 graphs :
(i) one graph from the figure below ;
(ii) one graph on your choice with about 10-12 vertices and about 20-24 edges ;
(iii) one larger biological graph.



For each of these graphs, you should compute its main parameters and analyse it. You can find in the tutorials the list of all parameters. But for each of the 3 graphs, you should at least compute the following :
(i) minimal, maximal, and average degrees ;
(ii) degree distribution ;
(iii) diameter, radius, and average path length ;
(iv) betweenness centrality ;
(v) Google PageRank ;
(v)i try to visualize the first two graphs.
You can also compute other parameters (and this will be taken into consideration in the evaluation).

2 How to organize your answer ?

In a .pdf file (which will be uploaded at the Ametice page of the course) you should explain what you have computed for all three graphs. Then for each of the graphs, you should provide your computations and the resulting values of the parameters. For some of the computations you should copy-past the Python execution, certifying your work.

3 Some indications and requirements.

(i) For the first two graphs, label the vertices of the graph.

(ii) To avoid that all people chose the same first graph, the students whose family name starts with one of the letters A-H analyze the first graphs from the figure, those with I-O analyze the second graph, and those with P-X the third graph.

(iii) For the third graph, you can try to upload it from some network repositories by searching Google using *biological graphs dataset*.